THE ADAMS SPECTRAL SEQUENCE FOR KO

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Abstract. This report provides an overview of the Adams spectral sequence, a powerful tool in algebraic topology for computing stable homotopy groups of spectra. We begin by introducing the foundational concepts of spectra and the Steenrod algebra. The construction of the Adams spectral sequence is then detailed, including the definition of Adams resolutions, the E_2-page in terms of Ext groups, and aspects of convergence and multiplicative structures. The report culminates in a discussion of the computation of the E_2-page of the Adams spectral sequence for the real connective K-theory spectrum, ko, outlining the key steps and the resulting calculation of $\pi_*(ko)^\wedge_2$.

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1. Introduction

Note: This report serves more as a summary of my current stage of learning. I have tried my best, within a limited timeframe, to understand as much as possible. Due to limitations in my background, I have not finished reading all the relevant materials, but I believe I have captured the main ideas. If a necessary concept is not fully understood, I still include it but I will point out the gap in my comprehension. I hope for your kind understanding.

As a starting point, we explain why one needs the category of spectrum. Briefly, if we take the objects to be merely based spaces (e.g. $\mathbf{Top}_*$), then one cannot give a satisfactory definition of morphisms: in invoking the Freudenthal suspension theorem one of the spaces must be a finite-dimensional CW complex. On the other hand, if we restrict our objects to finite CW complexes, then even the most complete construction—the Spanier–Whitehead stable homotopy category—fails to admit all coproducts. For these reasons, the necessity of passing to the category of spectrum is natural: spectra form a well-behaved, *stable* homotopy category in which these problems are resolved.

Definition 1.1. A *spectrum* is a sequence of based spaces $\{X_n\}_{n\geq 0}$ equipped with structure maps

$$\sigma_n : \Sigma X_n \longrightarrow X_{n+1},$$

which are adjoint to maps

$$\bar{\sigma}_n : X_n \longrightarrow \Omega X_{n+1}.$$

Every based space X can be naturally promoted to a spectrum, called the *suspension spectrum* $\Sigma^\infty X$, defined by

$$(\Sigma^\infty X)_n := \Sigma^n X$$

with structure maps given by the identity. From now on, we will often abuse notation by writing X to denote $\Sigma^\infty X$; this identification is natural and standard in the context of stable homotopy theory.

Definition 1.2. For each integer $n \in \mathbb{Z}$, the *n-sphere spectrum* S^n is defined as follows:

- For $n \geq 0$, S^n is the suspension spectrum of the based space S^n, with S^n placed in degree 0;
- For $n < 0$, S^n is the desuspension of S^0, i.e., $S^n = \Sigma^n S^0$, with S^0 placed in degree $-n$.

Remark 1.3. It is important to emphasize that when we write S, we always mean the sphere spectrum $\Sigma^\infty S^0$, not the suspension spectrum of S^1. These are fundamentally different objects in the category of spectrum.

We can define maps between spectra, as well as their homotopy classes, in order to obtain stable morphism groups between spectra. Due to space limitations, we will not develop these definitions in detail here.

Definition 1.4. The homotopy groups of a spectrum X are defined by

$$\pi_n(X) := [S^n, X],$$

In particular, $\pi_n(\Sigma^\infty X) = \pi_n^s(X)$ for X a based space.

As usual, one may define the *wedge sum* and the *smash product*. The wedge sum admits a fully general construction and poses no difficulties; however, when attempting to give a *strict* definition of the smash product within our current framework, one encounters serious limitations. In fact, the smash product can only be defined coherently in a *closed symmetric monoidal stable model category*.

I have to admit that I have not yet fully understand with the part about relevant categorical foundations. As such, the following discussion will proceed informally, working within a widely adopted modern framework such as the *category of symmetric spectra*, in which the smash product and other monoidal structures are already well-defined and rigorously treated.

Once the smash product has been defined, we may similarly define the *suspension* of a spectrum. In fact, this definition is fully compatible with what we have seen before in the case of based spaces.

Definition 1.5. Given a spectrum X, its n-fold suspension is defined by

$$\Sigma^n X := S^n \wedge X,$$

where S^n denotes the n-sphere spectrum.

It is important to observe that, once the smash product is available, the sphere spectrum S becomes the unit object for the monoidal structure; that is,

$$S \wedge X \cong X \cong X \wedge S.$$

Moreover, suspension and smash product interact compatibly. In particular, there are natural isomorphisms

$$\Sigma(X \wedge Y) \cong (\Sigma X) \wedge Y \cong X \wedge (\Sigma Y).$$

2. Steenrod alegbra

2.1. **Steenrod Operations.** In all that follows, unless otherwise noted, we work mod 2. Thus

$$H^*(X) = H^*(X; \mathbb{Z}/2), \qquad H_*(X) = H_*(X; \mathbb{Z}/2).$$

Definition 2.1. The Steenrod operations

$$\mathrm{Sq}^n \colon H^m(X) \longrightarrow H^{m+n}(X)$$

are characterized by the following five axioms:

(1) (*Naturality*) For every map $f \colon X \to Y$,
$$f^*(\mathrm{Sq}^n(x)) = \mathrm{Sq}^n\big(f^*(x)\big).$$
(2) (Sq^0 is the identity) $\mathrm{Sq}^0 = \mathrm{id} \colon H^m(X) \to H^m(X)$.
(3) For $x \in H^n(X)$, $\mathrm{Sq}^n(x) = x \smile x$.
(4) If $n > \deg(x)$ then $\mathrm{Sq}^n(x) = 0$.
(5) (*Cartan formula*) For all $x, y \in H^*(X)$,
$$\mathrm{Sq}^n(x \smile y) = \sum_{i+j=n} \mathrm{Sq}^i(x) \smile \mathrm{Sq}^j(y).$$

We will not give the topological construction of these operations, nor discuss their derivation. In what follows we will use only the normalization $\mathrm{Sq}^0 = \mathrm{id}$. We also note that Sq^1 is the Bockstein homomorphism and that it commutes with respect to suspension, the core of the entire Steenrod algebra lies in the following important **Adem relations**.

Proposition 2.2. *If $i < 2j$, then*

$$\mathrm{Sq}^i \mathrm{Sq}^j = \sum_{k=0}^{\lfloor i/2 \rfloor} \binom{j-k-1}{i-2k} \mathrm{Sq}^{i+j-k} \mathrm{Sq}^k.$$

This relation provides a mechanism for simplification: whenever a product $\mathrm{Sq}^i\mathrm{Sq}^j$ with $i < 2j$ appears, we may apply the Adem relation to rewrite it in terms of admissible monomials. We adopt the convention that $\binom{n}{k} = 0$ if $k < 0$ or $k > n$, so the relation, while perhaps inelegant, is combinatorially straightforward.

Example 2.3.

$$\begin{aligned}
&\mathrm{Sq}^1\mathrm{Sq}^1 = 0,\\
&\mathrm{Sq}^1\mathrm{Sq}^2 = \mathrm{Sq}^3,\\
&\mathrm{Sq}^2\mathrm{Sq}^3 = \mathrm{Sq}^5 + \mathrm{Sq}^4\mathrm{Sq}^1.
\end{aligned}$$

Definition 2.4. The mod-2 Steenrod algebra $\mathcal{A}$ is the graded, unital, associative algebra over $\mathbb{F}_2$ generated by Sq^i for $i \geq 0$, subject to $\mathrm{Sq}^0 = 1$ and the Adem relations.

For any based space X, the reduced cohomology $\widetilde{H}^*(X)$ is a left $\mathcal{A}$-module via

$$\mathrm{Sq}^I(x) := \mathrm{Sq}^{i_1} \cdots \mathrm{Sq}^{i_\ell}(x), \quad \lambda \colon \mathcal{A} \otimes \widetilde{H}^*(X) \to \widetilde{H}^*(X).$$

Definition 2.5. A sequence $I = (i_1, \ldots, i_\ell)$ of non-negative integers defines a product

$$\mathrm{Sq}^I := \mathrm{Sq}^{i_1} \cdots \mathrm{Sq}^{i_\ell} \in \mathcal{A}.$$

We say I is *admissible* if $i_s \geq 2i_{s+1}$ for all $1 \leq s < \ell$, with $i_\ell \geq 1$. The empty sequence $I = ()$ is admissible by convention, with length $\ell = 0$ and $\mathrm{Sq}^{()} = 1$.

We record the following two important theorems. The first is known as the *Milnor basis theorem.* No proof will be given here.

Theorem 2.6 (Milnor basis theorem)**.** *The admissible monomials form an $\mathbb{F}_2$-vector space basis of the Steenrod algebra:*

$$\mathcal{A} = \mathbb{F}_2\{\, \mathrm{Sq}^I \mid I \text{ admissible} \,\}.$$

Theorem 2.7. *The operation Sq^k is decomposable if and only if k is not a power of 2. Hence, the elements*

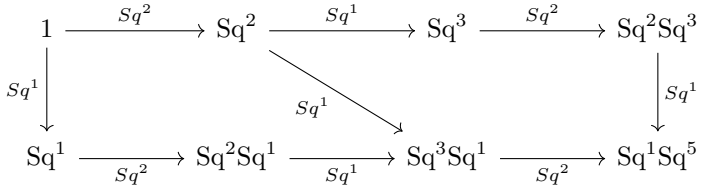
$$\mathrm{Sq}^{2^i}, \quad i \geq 0 \quad (\textit{e.g.}, \mathrm{Sq}^1, \mathrm{Sq}^2, \mathrm{Sq}^4, \mathrm{Sq}^8, \ldots)$$

generate $\mathcal{A}$ as an algebra.

We now mention a particular subalgebra of the Steenrod algebra that will play a special role at the end part of this report:

$$\mathcal{A}(1) := \langle \mathrm{Sq}^1, \mathrm{Sq}^2 \rangle \subset \mathcal{A}.$$

This is the subalgebra generated by Sq^1 and Sq^2. Below is an additive basis for $\mathcal{A}(1)$, with the actions of Sq^1 and Sq^2 indicated by arrows:



The Steenrod algebra is closely related, as an algebraic structure, to the cohomology of the mod-2 Eilenberg–MacLane spectrum.

2.2. **Eilenberg–MacLane spectrum.** To see this, we begin with the following definition:

Definition 2.8. Let $H = \{n \mapsto H_n := K(F_2, n)\}$ be the mod-2 Eilenberg–Mac Lane spectrum. The structure maps $\Sigma H_n \to H_{n+1}$ are adjoint to homotopy equivalences

$$H_n \simeq \Omega H_{n+1}, \quad \text{for all } n \geq 0.$$

We could define homology and cohomology of spectrum as follows or alternatively we could say it is a natural isomorphism if we start from somewhere else. One checks that $H^*(Y)$ carries a natural graded-ring structure (the cup product).

Definition 2.9. For any spectrum Y and integer n, set

$$H_n(Y) \;:=\; \pi_n\big(H \wedge Y\big) \;=\; [\, S^n,\; H \wedge Y \,], \quad H^n(Y) \;:=\; [\, Y,\; \Sigma^n H \,].$$

Proposition 2.10. *There is a canonical isomorphism of graded $\mathbb{F}_2$-algebras*

$$\mathcal{A} \cong H^*(H),$$

sending each Steenrod square Sq^i to its representing map $H \to \Sigma^i H$.

The main reason for introducing H is that for Y a spectrum bounded below (i.e. $\pi_*(Y)$ is bounded below) and with homology of finite type, then smashing with H yields a decomposition:

Proposition 2.11 (Splitting of $H \wedge Y$)**.** *Let Y be bounded below with $H_*(Y) = \mathbb{F}_2\{\alpha_u\}$ of finite type, and let $\{a_u\}$ be the dual basis of $H^*(Y)$, where $|a_u| = |\alpha_u| = n_u$. Denote by*

$$\alpha_u \colon S^{n_u} \longrightarrow H \wedge Y, \quad a_u \colon Y \longrightarrow \Sigma^{n_u} H$$

the representing maps. Then the canonical maps

$$\bigvee_u \Sigma^{n_u} H \xrightarrow{\simeq} H \wedge Y \quad \textit{and} \quad H \wedge Y \xrightarrow{\simeq} \prod_u \Sigma^{n_u} H$$

are stable equivalences.

In particular, for the most situations we apply relevant theory in practice, we usually are allowed to write

$$H \wedge Y \simeq \bigvee_u \Sigma^{n_u} H,$$

so that on cohomology

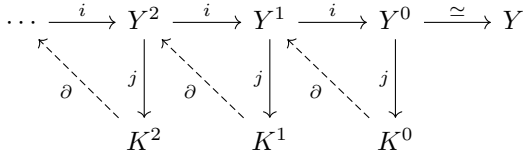
$$H^*(H \wedge Y) \cong \bigoplus_u H^*(\Sigma^{n_u} H) \cong \bigoplus_u \Sigma^{n_u} \mathcal{A},$$

acquires a natural $\mathcal{A}$-module structure. This splitting is essential in the construction of the Adams resolution.

3. The Adams spectral sequences

3.1. **Adams Resolutions.** In this section, unless otherwise stated, **all** spectrum are required to be bounded below and with homology of finite type. We now introduce the central notion of the Adams resolution.

Definition 3.1. An *Adams resolution* of spectrum Y is a diagram of spectrum



such that for each $s \geq 0$,

$$Y^{s+1} \xrightarrow{i} Y^s \xrightarrow{j} K^s \xrightarrow{\partial} \Sigma Y^{s+1}$$

is a cofiber sequence, and moreover:

(1) Each K^s has the form: $K^s \simeq \bigvee_u \Sigma^{n_u} H$.
(2) $j^* \colon H^*(K^s) \longrightarrow H^*(Y^s)$ is surjective.

We can always construct a *canonical Adams resolution* of a spectrum Y. Recall that in the stable homotopy category, the sphere spectrum S is the unit object. Let $f : S \to H$ be the unit map, and now we consider the cofiber sequence

$$\Sigma^{-1}\overline{H} \longrightarrow S \xrightarrow{f} H \longrightarrow \overline{H},$$

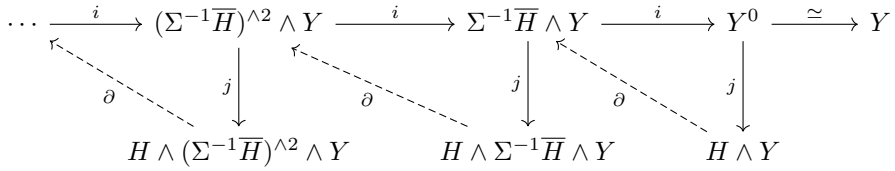
where $\overline{H} := \mathrm{cofiber}(f)$. Set

$$Y^0 = Y, \quad Y^{s+1} = \Sigma^{-1}\overline{H} \wedge Y^s, \quad K^s = H \wedge Y^s.$$

Then the cofiber sequence obtained by smashing Y^s:

$$Y^{s+1} \xrightarrow{i} Y^s \xrightarrow{j} K^s \xrightarrow{\partial} \Sigma Y^{s+1}$$

fits together into the following canonical Adams resolution:



Starting from the cofiber sequence we obtain above, do suspension:

$$\Sigma^s Y^{s+1} \xrightarrow{\Sigma^s i} \Sigma^s Y^s \xrightarrow{\Sigma^s j} \Sigma^s K^s \xrightarrow{\Sigma^s \partial} \Sigma^{s+1} Y^{s+1},$$

application of $H^*(-)$ yields a long exact sequence

$$\cdots \to H^{n-1}(\Sigma^s K^s) \xrightarrow{(\Sigma^s j)^*} H^{n-1}(\Sigma^s Y^s) \xrightarrow{(\Sigma^s i)^*} H^{n-1}(\Sigma^s Y^{s+1}) \xrightarrow{(\Sigma^s \partial)^*} H^n(\Sigma^s K^s) \to \cdots$$

We observe that the terms in LES (with replacements on notations):

$$H^{n-1}(\Sigma^s Y^{s+1}) \xrightarrow{\partial^*} H^n(\Sigma^s K^s) \xrightarrow{j^*} H^n(\Sigma^s Y^s).$$

where $H^{n-1}(\Sigma^s Y^{s+1}) = H^n(\Sigma^{s+1} Y^{s+1})$, will deduce a beautiful three–term exact sequence in degree n:

$$H^n(\Sigma^{s+1} Y^{s+1}) \xrightarrow{\partial^*} H^n(\Sigma^s K^s) \xrightarrow{j^*} H^n(\Sigma^s Y^s).$$

From the above process, we will obtain a nice resolution, which is:

$$\cdots \longrightarrow P_s \xrightarrow{\partial_s} P_{s-1} \longrightarrow \cdots \xrightarrow{\partial_1} P_0 \xrightarrow{\epsilon} H^*(Y) \longrightarrow 0$$

It is a resolution of $H^*(Y)$ by **free $\mathcal{A}$-modules**, each of which is bounded below and of finite type, and

$$P_s := H^*(\Sigma^s K^s), \quad \partial_s := \partial^* \circ j^* \colon H^*(\Sigma^s K^s) \to H^*(\Sigma^{s-1} K^{s-1}), \quad \epsilon := j^* \colon H^*(K^0) \longrightarrow H^*(Y).$$

To see the free $\mathcal{A}$-modules, only need to notice that $K^s = H \wedge Y^s$ for Y is assumed to be bounded below and with homology of finite type so we can apply the splitting property mentioned above and obtain the structure we want.

Since we have constructed a free resolution, it is now natural to consider the Ext functor.

3.2. **The Adams spectral sequences of E_2 page.** We start this subsection from recalling some basic definitions in spectral sequences.

Definition 3.2. A *spectral sequence of Adams type* is a sequence of bigraded abelian groups

$$E_r^{*,*} = \{E_r^{s,t}\}_{s,t},$$

together with differentials

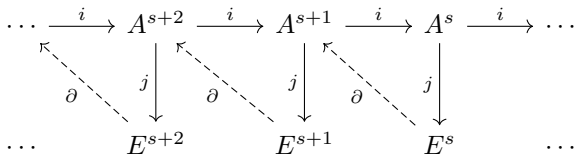
$$d_r \colon E_r^{*,*} \to E_r^{*,*}$$

of bidegree $(r, r-1)$, and isomorphisms between pages

$$E_{r+1}^{s,t} \cong H^{s,t}(E_r^{*,*}, d_r) \quad \text{for all } r \geq 1.$$

We call $E_r^{*,*}$ the E_r-term, d_r the d_r-differential, s the filtration degree, and $t-s$ the total degree of the spectral sequence.

Definition 3.3. An *unrolled exact couple of Adams type* is a diagram of graded abelian groups and homomorphisms:



such that each triangle

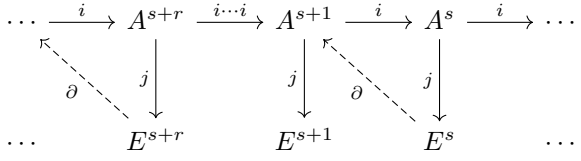
$$\cdots \longrightarrow A^{s+1} \xrightarrow{i} A^s \xrightarrow{j} E^s \xrightarrow{\partial} A^{s+1} \longrightarrow \cdots$$

is a long exact sequence. And in the Adams case, the bidegrees of the maps i,j,∂ are: $(-1,-1)$, $(0,0)$, $(1,0)$, which will be explained later in a case in details.

Definition 3.4. The r-th ($r \geq 1$) cocycle subgroup of E^s is defined to be

$$Z_r^s := \partial^{-1}\big(\operatorname{im}(i^{r-1} \colon A^{s+r} \to A^{s+1})\big).$$

To see this, we consider the diagram:



This should make sense, and similarly we can define:

Definition 3.5. The r-th ($r \geq 1$) boundary subgroup of E^s is defined to be

$$B_r^s := j\big(\ker(i^{r-1} \colon A^s \to A^{s-r+1})\big).$$

Definition 3.6. The filtration degree s component of the E_r-term is defined to be

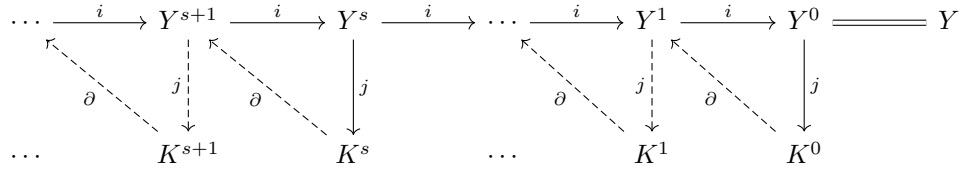
$$E_r^s := Z_r^s / B_r^s$$

Remark 3.7. Clearly $Z_1^s = E^s$ and $B_1^s = 0$, so $E_1^s = E^s$.

From here, we can define the r-th differential $d_r^s\colon E_r^s \longrightarrow E_r^{s+r}$ by the rule $d_r^s([x]) = [j(y)]$, where $x \in Z_r^s$, $y \in A^{s+r}$, and $\partial(x) = i^{r-1}(y)$. Such process should also make sense in the above diagram.

We can check the fact that d_r is well-defined, and $\ker d_r^s \cong Z_{r+1}^s/B_r^s, \operatorname{im} d_r^s = B_{r+1}^s/B_r^s$, so that $H^s(E_r^{*,*}, d_r) \cong E_{r+1}^s$. Hence $\{E_r, d_r\}_r$ is a spectral sequence of Adams type while d_r has bidegree $(r, r-1)$ in this case.

Proposition 3.8. *There is a very important and basic result for the tower of spectrum in the following type:*



where K^s is the mapping cone of $i : Y^{s+1} \to Y^s$, and $\partial : K^s \to \Sigma Y^{s+1}$ is the cofiber map.

Applying homotopy will yield an unrolled exact couple of Adams type, giving rise to a spectral sequence of Adams type with E_1-term

$$E_1^{s,t} = \pi_{t-s}(K^s).$$

And for $s \geq 0$, the $d_1^{s,t}$-differential

$$d_1^{s,t}\colon E_1^{s,t} \longrightarrow E_1^{s+1,t}$$

is given by the composite

$$\pi_{t-s}(K^s) \xrightarrow{\partial} \pi_{t-s-1}(Y^{s+1}) \xrightarrow{j_*} \pi_{t-s-1}(K^{s+1}).$$

Here we can see that for the diagram

$$\begin{array}{ccc} \pi_{t-s}(Y^s) & \xrightarrow{\quad i \quad} & \pi_{(t-1)-(s-1)}(Y^{s-1}) \\ & & \downarrow{\scriptstyle j} \\ \pi_{(t-1)-s}(Y^s) & \xleftarrow{\quad \partial \quad} & \pi_{(t-1)-(s-1)}(K^{s-1}) \end{array}$$

Clearly explain the meaning of bidegrees of the maps i,j,∂ are: $(-1,-1)$, $(0,0)$, $(1,0)$.

Applying this proposition to the Adams resolution of Y, we obtain the E_1-term

$$E_1^{s,t} \;=\; \pi_{t-s}(K^s),$$

and hence the chain complex

$$\cdots \xleftarrow{(j\partial)_*} \pi_*(\Sigma^2 K^2) \xleftarrow{(j\partial)_*} \pi_*(\Sigma K^1) \xleftarrow{(j\partial)_*} \pi_*(K^0) \leftarrow 0.$$

where we are using the suspension identification $\pi_*(\Sigma^s K^s) \cong \pi_{*-s}(K^s)$,

From class we shall understand the following complex is achieved under d-invariant.

$$\cdots \longleftarrow \operatorname{Hom}_{\mathcal{A}}(H^*(\Sigma^2 K^2), \mathbb{F}_2) \xleftarrow{((j\partial)^*)^*} \operatorname{Hom}_{\mathcal{A}}(H^*(\Sigma K^1), \mathbb{F}_2) \xleftarrow{((j\partial)^*)^*} \operatorname{Hom}_{\mathcal{A}}(H^*(K^0), \mathbb{F}_2) \longleftarrow 0$$

which is just

$$\cdots \longleftarrow \mathrm{Hom}_{\mathcal{A}}(P_2, \mathbb{F}_2) \xleftarrow{\partial_2^*} \mathrm{Hom}_{\mathcal{A}}(P_1, \mathbb{F}_2) \xleftarrow{\partial_1^*} \mathrm{Hom}_{\mathcal{A}}(P_0, \mathbb{F}_2) \longleftarrow 0.$$

Now we have strongly relate the complex deduced by E_1-term and the free resolution we constructed from adams resolution, thus take their cohomology will deduce the isomorphic result, which happens to be E_2-term and the Ext-term thus we can deduce the Adams spectral sequence takes the form:

$$E_2^s \cong \mathrm{Ext}_{\mathcal{A}}^s(H^*(Y), \mathbb{F}_2)$$

Now we discuss the internal grading. Note that

$$E_1^{s,t} = \pi_{t-s}(K^s) = [S^{t-s},\, K^s] \;\cong\; \mathrm{Hom}_{\mathcal{A}}\big(H^*(\Sigma^s K^s),\, \Sigma^t H^*(S)\big) = \mathrm{Hom}_{\mathcal{A}}\big(H^*(\Sigma^s K^s),\, \Sigma^t\, \mathbb{F}_2\big).$$

Equivalently, this is the group of $\mathcal{A}$-module homomorphisms

$$H^*(\Sigma^s K^s) \;\longrightarrow\; \Sigma^t\, \mathbb{F}_2$$

that lower degree by t. We denote it by

$$\mathrm{Hom}_{\mathcal{A}}^t\big(H^*(\Sigma^s K^s),\, \mathbb{F}_2\big).$$

Hence

$$E_2^{s,t} = \mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(Y),\, \mathbb{F}_2\big).$$

From above, we have shown the main theorem of the whole Adams spectral sequence theory, that is:

Theorem 3.9. *The E_2-term of the Adams spectral sequence for Y is*

$$E_2^{s,t} \;=\; \mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(Y),\, \mathbb{F}_2\big).$$

which also gives that this is independent of the choice of Adams resolution.

A natural extension gives:

Theorem 3.10. *The E_2-term of the Adams spectral sequence $\{E_r(X,Y) = E_r^{*,*}(X,Y)\}_r$ of maps $X \to Y$ is*

$$E_2^{s,t} = \mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(Y), H^*(X)).$$

3.3. **Convergence.**

Definition 3.11. Let $\{E_r^{s,t}\}_{r\geq 1}$ be a spectral sequence of Adams type, and let G_* be a graded abelian group. We say that

$$E_r^{s,t} \;\Longrightarrow\; G_{\,t-s}$$

(or that $\{E_r\}$ *converges* or *abuts* to G_*) if for each total degree $n = t - s$ there is a finite, exhaustive, and separated filtration

$$0 = F^{N+1}G_n \;\subseteq\; F^N G_n \;\subseteq\; \cdots \;\subseteq\; F^0 G_n = G_n$$

such that the associated graded pieces satisfing

$$\mathrm{gr}_F^s\, G_n \;=\; F^s G_n / F^{s+1} G_n \;\cong\; E_\infty^{s,\,s+n},$$

where

$$E_\infty^{s,t} \;=\; \lim_{r\to\infty} E_r^{s,t}.$$

Here we directly state the main theorem of this part, which is also an example of convergence.

Theorem 3.12. *For the Adams spectral sequence of a spectrum Y, one shows that it converges to the 2-adic completion of its stable homotopy groups. In our notation,*

$$E_2^{s,t} \;=\; \mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(Y), \mathbb{F}_2\big) \quad \Longrightarrow \quad \pi_{t-s}(Y)^{\wedge}_2.$$

So here we also need to mention the definition of p-adic completion.

Definition 3.13. Let A be an abelian group and let p be a prime. The *p-adic completion* of A is the inverse limit

$$\widehat{A}_p \;:=\; \varprojlim_n A/p^nA,$$

equipped with the inverse limit topology. Concretely, this is analogous to the construction of the p-adic integers $\mathbb{Z}_p = \varprojlim_n \mathbb{Z}/p^n\mathbb{Z}$.

I won't (in fact I can't) devote too much space to a detailed discussion of this part, including the proof of the theorem and analysis of related properties. This is not because I consider this content unimportant—on the contrary, I believe this part constitutes the algebraic core of every spectral sequence discussion and is of fundamental significance.

To be fully transparent, I have not yet developed a deep understanding of this part, and I do not wish to write content that I do not fully grasp. We will instead give a very rough definition and a preliminary analysis as follows as the end of this subsection:

Definition 3.14. Let $\pi_n(Y)^{\wedge}_p = \varprojlim_e \pi_n(Y)/p^e$ be the algebraic p-completion of $\pi_n(Y)$.

Specially, if $\pi_*(Y)$ is of finite type, i.e., if $\pi_n(Y)$ is finitely generated for each n, then

$$\pi_n(Y)\otimes\mathbb{Z}_p \;\cong\; \pi_n(Y)^{\wedge}_p \;\cong\; \pi_n(Y^{\wedge}_p) \quad \text{for all } n.$$

where the notation $Y^{\wedge}_p$ is the p-completion of a spectrum, given by the following definition:

Definition 3.15. Fix a prime p. For a spectrum Y, consider the tower of cofiber sequences

$$Y \;\longrightarrow\; Y\wedge S/p^e \;\longrightarrow\; Y\wedge S/p^{e-1} \;\longrightarrow\; \cdots \;\longrightarrow\; Y\wedge S/p,$$

where S/p^e is the cofiber of $p^e\colon S \to S$. The *p-completion* of Y is defined to be the homotopy limit

$$Y^{\wedge}_p \;:=\; \mathrm{holim}_e\,\big(Y\wedge S/p^e\big).$$

The canonical maps $Y \to Y\wedge S/p^e$ then assemble to the p-completion map $Y \to Y^{\wedge}_p$.

3.4. **Naturality and Multiplicative Structure.**

Theorem 3.16. *Let $f\colon Y \to Y'$ be a map of connective spectra i.e. $\pi_n(Y) = 0$ for all $n < 0$, and fix any spectrum X. Then f induces a morphism of Adams spectral sequences*

$$f_*\colon \{E_r^{s,t}(X,Y),\, d_r\} \;\longrightarrow\; \{E_r^{s,t}(X,Y'),\, d_r\},$$

which on the E_2-page is given by

$$(f^*)^*\colon\; \mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(Y'),\, H^*(X)\big) \;\longrightarrow\; \mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(Y),\, H^*(X)\big),$$

where $f^*\colon H^*(Y') \to H^*(Y)$ is the induced $\mathcal{A}$-module homomorphism.

Proof Idea. The full proof is quite not short, but its essential step is the construction of a chain map

$$F_\bullet \colon P_\bullet(Y) \longrightarrow P_\bullet(Y')$$

lifting the given map $f \colon Y \to Y'$, which is unique up to chain homotopy. This chain map induces the morphism of cochain complexes $\mathrm{Hom}(P_\bullet(Y'), \mathbb{F}_2) \to \mathrm{Hom}(P_\bullet(Y), \mathbb{F}_2)$ and hence the map of Adams spectral sequences. We omit the detailed arguments and content ourselves with this outline.

Now we give the definition of pairing to start the multiplicative structure part.

Definition 3.17. Let $\{E'_r\}$, $\{E''_r\}$ and $\{E_r\}$ be three spectral sequences (of Adams type), each with bigraded terms $E_r^{*,*}$ and differentials d_r. A *pairing* of these spectral sequences is a collection of bilinear maps

$$\phi_r \colon {E'_r}^{s,t} \otimes {E''_r}^{s',t'} \longrightarrow {E_r}^{s+s',\, t+t'}$$

for all $r \geq 1$, satisfying the following conditions:

(1) For homogeneous elements $x \in E'_r$ and $y \in E''_r$,

$$d_r\big(\phi_r(x \otimes y)\big) = \phi_r\big(d_r(x) \otimes y\big) \;+\; (-1)^{|x|}\, \phi_r\big(x \otimes d_r(y)\big).$$

(2) If $x \in \ker d_r \subset E'_r$ and $y \in \ker d_r \subset E''_r$, let $[x] \in E'_{r+1}$ and $[y] \in E''_{r+1}$ denote their homology classes. Then

$$\phi_{r+1}\big([x] \otimes [y]\big) \;=\; \big[\phi_r(x \otimes y)\big] \quad \in E_{r+1}.$$

An interesting aspect of pairings in the Adams spectral sequence is that we can define two types of pairings:

- The classical *tensor product* pairing.
- The *composition* pairing, which is especially notable because it corresponds, on the E_2-page, to the **Yoneda pairing**.

Definition 3.18. Let $\mathcal{A}$ be a graded algebra and let M, N, P be graded $\mathcal{A}$-modules. For integers $s, t \geq 0$, the *Yoneda pairing* is the bilinear map

$$\cup \colon \mathrm{Ext}^s_{\mathcal{A}}(M, N) \otimes \mathrm{Ext}^t_{\mathcal{A}}(N, P) \longrightarrow \mathrm{Ext}^{s+t}_{\mathcal{A}}(M, P)$$

defined by "splicing" extensions. Concretely, if

$$0 \to N \to E_s \to \cdots \to E_1 \to M \to 0 \quad \text{represents } [x] \in \mathrm{Ext}^s_{\mathcal{A}}(M, N),$$

and

$$0 \to P \to F_t \to \cdots \to F_1 \to N \to 0 \quad \text{represents } [y] \in \mathrm{Ext}^t_{\mathcal{A}}(N, P),$$

then

$$[x] \cup [y] \;=\; \Big[0 \to P \to F_t \to \cdots \to F_1 \to E_s \to \cdots \to E_1 \to M \to 0\Big],$$

which defines an element of $\mathrm{Ext}^{s+t}_{\mathcal{A}}(M, P)$. This pairing coincides with the composition product in the derived category of $\mathcal{A}$-modules.

The following two theorems shows two sorts of pairings:

Theorem 3.19 (Given by Adams)**.** *Let X and Y be spectra. For each $r \geq 1$ there is a natural pairing of Adams spectral sequences*

$$E_r^{s,t}(X) \otimes E_r^{u,v}(Y) \xrightarrow{\otimes} E_r^{s+u,\, t+v}(X \wedge Y),$$

which on the E_2-page is induced by the tensor-product pairing

$$\mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(X),\mathbb{F}_2\big)\ \otimes\ \mathrm{Ext}_{\mathcal{A}}^{u,v}\big(H^*(Y),\mathbb{F}_2\big)\ \longrightarrow\ \mathrm{Ext}_{\mathcal{A}}^{s+u,\,t+v}\big(H^*(X\wedge Y),\mathbb{F}_2\big),$$

and which converges to the smash-product pairing

$$\pi_{t-s}(X_2^\wedge)\ \otimes\ \pi_{v-u}(Y_2^\wedge)\ \longrightarrow\ \pi_{\,t+v-s-u}((X\wedge Y)_2^\wedge.)$$

Theorem 3.20 (Given by Moss)**.** *Let X, Y, and Z be spectra. For each $r\geq 1$ there is a natural pairing of Adams spectral sequences*

$$\circ:\ E_r^{s,t}(Y,Z)\ \otimes\ E_r^{u,v}(X,Y)\ \longrightarrow\ E_r^{s+u,\,t+v}(X,Z),$$

which on the E_2-page agrees with the Yoneda composition

$$\mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(Z),H^*(Y)\big)\otimes\mathrm{Ext}_{\mathcal{A}}^{u,v}\big(H^*(Y),H^*(X)\big)\ \longrightarrow\ \mathrm{Ext}_{\mathcal{A}}^{s+u,\,t+v}\big(H^*(Z),H^*(X)\big),$$

and which converges to the usual composition pairing on completed homotopy groups

$$[Y,Z_p^\wedge]_*\ \otimes\ [X,Y_p^\wedge]_*\ \longrightarrow\ [X,Z_p^\wedge]_*.$$

Moreover, this pairing is associative and unital.

Remark 3.21. In the special case $X=S^0$, the Adams spectral sequence for any spectrum Y becomes a module over the sphere's E_2–term:

$$E_2^{*,*}(S)\ \otimes\ E_2^{*,*}(Y)\ \longrightarrow\ E_2^{*,*}(Y),$$

where

$$E_2^{*,*}(S)=\mathrm{Ext}_{\mathcal{A}}^{*,*}(\mathbb{F}_2,\mathbb{F}_2).$$

This action is most conveniently visualized on the E_2–page by drawing short line segments between those classes in $E_2^{s,t}(Y)$ which are related by multiplication by generators of $E_2^{*,*}(S)$. For example:

a vertical line from $(s,t)\to(s+1,t+1)$ indicates multiplication by $h_0\in E_2^{1,1}(S)$,

a diagonal line of slope 1 from $(s,t)\to(s+1,t+2)$ indicates multiplication by $h_1\in E_2^{1,2}(S)$.

These connecting segments are *not* differentials but rather the graphical appearance of the pairing with the sphere. Such annotations make the underlying multiplicative structure of the E_2–page immediately apparent.

4. E_2-page of the Adams spectral sequence for ko

In this section we turn to an explicit computation of the E_2-page of the Adams spectral sequence converging to $\pi_*(ko)_2^\wedge$. By definition,

$$E_2^{s,t}\ =\ \mathrm{Ext}_{\mathcal{A}}^{s,t}\big(H^*(ko),\,\mathbb{F}_2\big).$$

So our first task is to find what is $H^*(ko)$.

In fact, the following result from Stong clearly shows that

Proposition 4.1. *There is an $\mathcal{A}$-module isomorphism*

$$H^*(ko)\ \cong\ \mathcal{A}\big/A(1).$$

We assume this result as a known fact and now we state another useful result to help us with computations:

Proposition 4.2.

$$\mathrm{Ext}_{\mathcal{A}}^{*,*}(\mathcal{A}/\mathcal{A}(1),\mathbb{F}_2)\ \cong\ \mathrm{Ext}_{\mathcal{A}(1)}^{*,*}(\mathbb{F}_2,\mathbb{F}_2).$$

Proof Sketch. Write each admissible monomial of the Steenrod algebra $\mathcal{A}$ in Milnor form $\mathrm{Sq}^{i_1}\cdots\mathrm{Sq}^{i_\ell}$ with $i_1 \geq \cdots \geq i_\ell$. For every exponent i_s express $i_s = 4q_s + r_s$ with $0 \leq r_s < 4$. Then now every Milnor monomial factors uniquely as

$$\alpha\beta, \quad \alpha \in \mathcal{A}(1),\ \beta \in P,$$

where $P := \mathbb{F}_2\{\text{admissible } \mathrm{Sq}^{4q_1}\mathrm{Sq}^{4q_2}\cdots\}$ by collecting the factors with $r_s < 4$ into α and the factors divisible by 4 into β.

Now we can see the assignment $\alpha \otimes \beta \mapsto \alpha\beta$ yields a vector space bijection

$$\mathcal{A}(1) \otimes_{\mathbb{F}_2} P \xrightarrow{\cong} \mathcal{A},$$

which respects the right $\mathcal{A}(1)$-action. Hence $\mathcal{A}$ is *right free* (therefore flat) over $\mathcal{A}(1)$.

Since $\mathcal{A}$ is right flat over $\mathcal{A}(1)$, it should(and will) gives

$$\mathrm{Ext}^{*,*}_{\mathcal{A}}\big(\mathcal{A}/\mathcal{A}(1),\, \mathbb{F}_2\big) \;\cong\; \mathrm{Ext}^{*,*}_{\mathcal{A}(1)}\big(\mathbb{F}_2,\, \mathbb{F}_2\big),$$

from some algebra knowledge which completes the proof. □

Now we will show how to find a possible resolution, and we begin with the chain complex

$$\cdots \longrightarrow P_s \xrightarrow{\partial_s} P_{s-1} \longrightarrow \cdots \xrightarrow{\partial_1} P_0 \xrightarrow{\epsilon} \mathbb{F}_2 \longrightarrow 0.$$

Set

$$P_0 = \mathcal{A}(1)\langle A_0\rangle,$$

where A_0 is a formal generator and $\epsilon\colon P_0 \to \mathbb{F}_2$ is defined by $\epsilon(A_0) = 1$. Hence

$$\ker(\epsilon) = \mathrm{Span}_{\mathbb{F}_2}\big\{\mathrm{Sq}^I A_0 \mid I \neq ()\big\},$$

and the seven generators of $\ker(\epsilon)$ are

$$\mathrm{Sq}^1A_0,\ \mathrm{Sq}^2A_0,\ \mathrm{Sq}^2\mathrm{Sq}^1A_0,\ \mathrm{Sq}^3A_0,\ \mathrm{Sq}^3\mathrm{Sq}^1A_0,\ \mathrm{Sq}^2\mathrm{Sq}^1\mathrm{Sq}^2A_0,\ \mathrm{Sq}^5\mathrm{Sq}^1A_0.$$

To surject onto $\ker(\epsilon)$, we define

$$P_1 = \mathcal{A}(1)\langle B_1, B_2\rangle,$$

with $\partial_1(B_1) = \mathrm{Sq}^1A_0$ and $\partial_1(B_2) = \mathrm{Sq}^2A_0$. Since $\partial_1(\mathrm{Sq}^1B_1) = \mathrm{Sq}^1\mathrm{Sq}^1A_0 = 0$, the class Sq^1B_1 lies in $\ker\partial_1$, and similarly Sq^2B_1 covers $\mathrm{Sq}^2\mathrm{Sq}^1A_0$. By choosing B_2 to hit Sq^2A_0, one checks that the $\mathcal{A}(1)$–orbits of B_1, B_2 indeed generate the entire kernel.

Target in P_0	Preimages in P_1
Sq^1A_0	B_1
Sq^2A_0	B_2
$\mathrm{Sq}^{2,1}A_0$	Sq^2B_1
Sq^3A_0	Sq^1B_2
$\mathrm{Sq}^{3,1}A_0$	$\mathrm{Sq}^3B_1,\ \mathrm{Sq}^2B_2$
$\mathrm{Sq}^{2,1,2}A_0$	$\mathrm{Sq}^{2,1}B_2$
$\mathrm{Sq}^{5,1}A_0$	$\mathrm{Sq}^{2,1,2}B_1,\ \mathrm{Sq}^{3,1}B_2$
0	$\mathrm{Sq}^1B_1,\ \mathrm{Sq}^{2,1}B_1,\ \mathrm{Sq}^{3,1}B_1,$ $\mathrm{Sq}^3B_2,\ \mathrm{Sq}^{5,1}B_1,\ \mathrm{Sq}^{2,1,2}B_2,\ \mathrm{Sq}^{5,1}B_2$

Thus $P_1 = \mathcal{A}(1)\langle B_1, B_2\rangle$ surjects onto $\ker(\epsilon)$, completing the first step of the resolution. Repeatly, we can get all of them.

We observe that the construction of the entire resolution is, in essence, not particularly complicated, but it requires substantial patience to carry out the step-by-step computations. While we do not provide the full construction of the resolution here, we emphasize that it is, in principle, **completely computable**.

Now we start to **explain** how to compute the E_2-page in general, consider the dual cochain complex

$$\cdots \longrightarrow \mathrm{Hom}_{\mathcal{A}}(P_1, \mathbb{F}_2) \xrightarrow{\partial_1^*} \mathrm{Hom}_{\mathcal{A}}(P_0, \mathbb{F}_2) \longrightarrow 0,$$

we could pick the *dual basis* $\{A_0^*\}$ for $\mathrm{Hom}_{\mathcal{A}}(P_0, \mathbb{F}_2)$ and $\{B_1^*, B_2^*, \dots\}$ for $\mathrm{Hom}_{\mathcal{A}}(P_1, \mathbb{F}_2)$.

By definition, $\partial_1^*(A_0^*)$ is the linear combination of the B_i^* given by the coefficients in $\partial_1(B_i)$.Then we can compute $\ker(\partial_1^*), \mathrm{im}(\partial_1^*)$ in each internal degree t, and get the cohomology groups we want which are exactly $\mathrm{Ext}_{\mathcal{A}}^{s,t}(H^*(Y), \mathbb{F}_2)$.

Finally, we should get a beautiful diagram from finite steps of calculations:

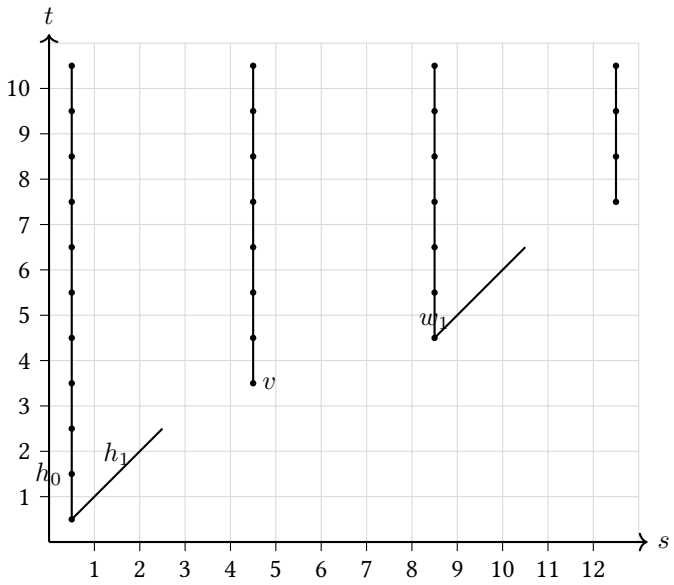


FIGURE 1. Adams E_2-page for ko with key classes labeled: h_0 at $(s,t) = (0,1)$, h_1 at $(1,1)$, v at $(4,3)$, and w_1 at $(8,4)$.

For the above chart, it is easy to see that the only potentially nontrivial differential is the one from column $8k+1$ to column $8k$, where $k \in \mathbb{N}$.

We can verify that this differential vanishes using the derivation rule and computations among the generators. Hence, we conclude that

$$E_2^{s,t}(ko) \;=\; E_\infty^{s,t}(ko).$$

Finally we would deduce our final result via reading the page, that is:

$$\pi_*(ko_2^\wedge) = \mathbb{Z}_2[\eta, \alpha, \beta]/(2\eta,\ \eta^3,\ \eta\alpha,\ \alpha^2 = 4\beta).$$

The classes 2, η, α, *and* β *are represented by* h_0, h_1, v, w_1 *respectively at the diagram above.*

5. Conclusion

This concludes the main content of the report. Thank you for reading.

As a summary, I do not think it is necessary to review what I have done. Instead, I would like to list a few shortcomings that I am aware of. Due to the limitations of my own ability, I must acknowledge the following aspects where my work is insufficient:

(1) Initially, I believed that categorical questions related to the definition of spectra were not particularly difficult. However, once I attempted to write things down clearly, I realized that the category of spectra contains many deep and subtle issues. This is an area I intend (and need) to study further.
(2) I did not provide detailed algebraic proofs. The entire report reads more like a study summary, consisting mostly of my discussions of definitions and my understanding of proof sketches.
(3) I admit that I do not yet have a clear understanding of the convergence part, and in fact, the section on p-completion is also something I have not fully grasped. Nevertheless, I chose to include them in order to preserve completeness. I ask for your understanding.
(4) Due to poor time management, I originally thought I had sufficient time to write about the computation of ko, but I found myself running out of time. As a result, the final part of the report contains many gaps. I apologize for that.